



University
of Glasgow

W10-02: Legal and Ethical Considerations

COMPSCI4064 (H) and COMPSCI5088 (M)

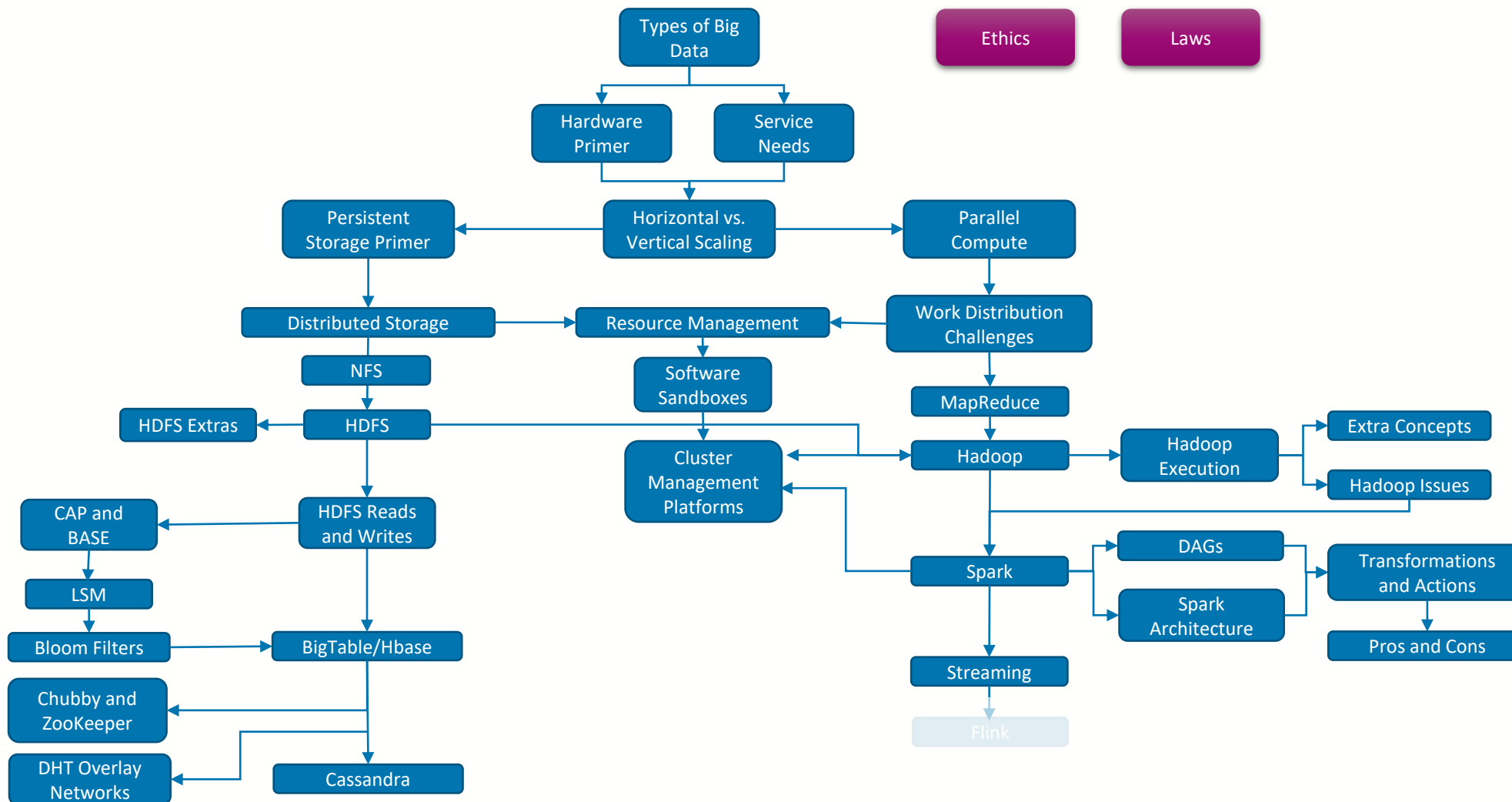
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Where are we?



With “big” data comes “big” responsibility...

- Big Data systems are powerful as they can process and analyse large volumes of data very quickly
 - Often used to identify patterns of behaviour that have business value
 - However, this can be harmful to individuals or society as a whole
- So we need to be mindful of where we should not use the Big Data available to us



Consider this example...

- You are working at a political consulting company, who has just completed an online survey of around 250,000 Facebook (Meta) users
- When looking at the data, you find that beyond the 250k user survey, there is also attached information about the social network of each user
 - E.g. public profile, page likes, birthday and current city of their friends
- Is it **ethical** and or **legal** to process this data?

No!



Facebook–Cambridge Analytica Scandal

- In 2013, a political consulting firm Cambridge Analytica developed a Facebook (Meta) app called *This Is Your Digital Life*
 - Facebook users could take a survey for a fee
 - They were informed that the survey was for academic use, and it was developed by the University of Cambridge
- Beyond collecting the survey data, it used the Facebook API to crawl information about each user's friend network
 - In total it collected data for 87 million users
 - The data was detailed enough for Cambridge Analytica to create psychographic profiles of the subjects of the data
 - This data was then used for targeted political campaigning - including the U.S. Trump presidential campaign and the UK's Vote Leave campaign
- When this was made public, \$100 billion was knocked off Facebook's market capitalization in days
- Cambridge Analytica filed for bankruptcy
- Facebook was fined \$5 billion



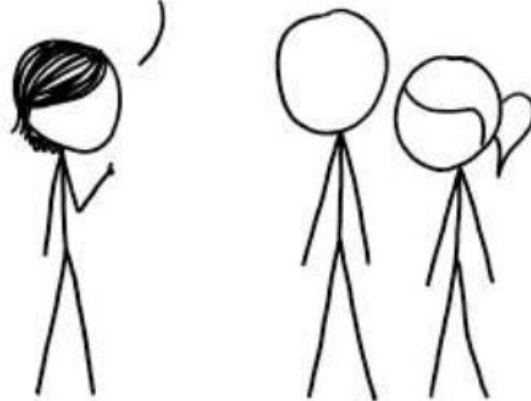
Big Data Ethics

FACEBOOK SHOULDN'T CHOOSE WHAT STUFF THEY SHOW US TO CONDUCT UNETHICAL PSYCHOLOGICAL RESEARCH.

THEY SHOULD ONLY MAKE THOSE DECISIONS BASED ON, UH...

HOWEVER THEY WERE DOING IT BEFORE.

WHICH WAS PROBABLY ETHICAL, RIGHT?

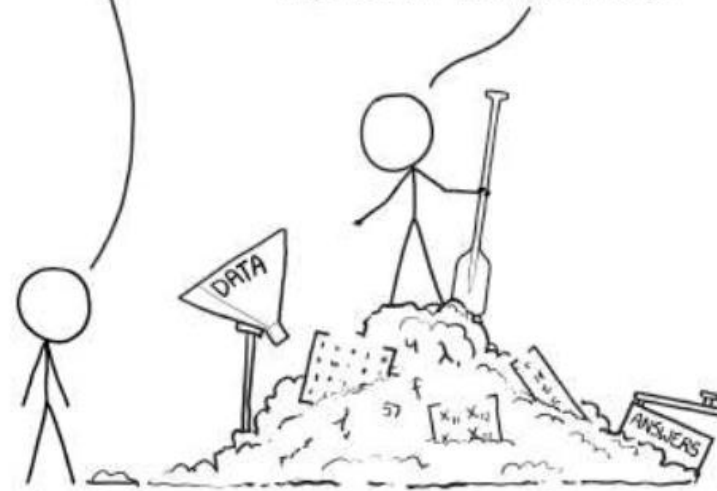


THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG PILE OF LINEAR ALGEBRA, THEN COLLECT THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL THEY START LOOKING RIGHT.





Core Concepts: Privacy

- Privacy is recognised as a human right under numerous declarations and treaties. In the UK, the ECHR has been implemented through the Human Rights Act 1998, with protection of personal data provided by the Data Protection Act 1998
- The definition is somewhat fuzzy though – and humans tend not to understand the implications when they are publishing publicly online
 - Are social media spaces public or private?
 - Important to unambiguously distinguish public and private users or usages – but difficult in practice



Core Concepts: Informed Consent

- The idea that an individual consents to the processing of their data for clearly understood uses
- But... in big data analytics, very little may be known about intended future uses of data when it is collected
 - It is unlikely that consent obtained at the point of data collection (one-off) would meet a strict definition of “informed consent”.
 - The validity of consent obtained by agreement to terms and conditions is debateable
 - Think of all of those terms and conditions you don’t read
 - Obtaining informed consent may be impossible or prohibitively expensive



Core Concept: De-identification

- De-identification/anonymization/pseudonymization is the **process** where data elements that might immediately **identify a person or organisation** have been **removed** or **masked**.
- However, recognition is growing that such distinctions are becoming less tenable
 - Identifiability is increasingly being seen as a continuum, not binary.
 - Disclosure risks increase with dimensionality (i.e., number of variables), linkage of multiple data sources, and the power of data analytics.
 - Disclosure risks can be mitigated, but not eliminated.



Core Concept: Research Integrity

- Data repositories play a vital role in supporting research integrity by holding data and making them available to others for both validation, replication, as well as providing new research opportunities.
 - Data must have clear “**provenance**”, its sources and processing need to be known, identified, and documented.
- But this can be tricky:
 - Much data not collected for research, such as administrative data, has different standards (e.g., quality, metadata) to research data.
 - For some genres, often with commercial value, such as Twitter data, there are legal restrictions on reproducing data



Laws: Data Protection Act 1998

- Designed to protect "personal data"
 - Any data that could have been used to identify a living individual.
 - An individual has to consent to the collection of their personal information
- Data protection principles
 1. Personal data shall be processed fairly and lawfully and, in particular, shall not be processed unless
 - Necessary for the performance of, or commencing, a contract
 - Required under a legal obligation
 2. Personal data shall be obtained only for one or more specified and lawful purposes, and shall not be further processed in any manner incompatible with that purpose or those purposes.
 3. Personal data shall be adequate, relevant and not excessive in relation to the purpose or purposes for which they are processed.
 4. Personal data shall be accurate and, where necessary, kept up to date.
 5. Personal data processed for any purpose or purposes shall not be kept for longer than is necessary for that purpose or those purposes.
 6. Personal data shall be processed in accordance with the rights of data subjects (individuals).
 7. Appropriate technical and organizational measures shall be taken against unauthorised or unlawful processing of personal data and against accidental loss or destruction of, or damage to, personal data.
 8. Personal data shall not be transferred to a country or territory outside the European Economic Area unless that country or territory ensures an adequate level of protection for the rights and freedoms of data subjects in relation to the processing of personal data.



Laws:

Freedom of Information Act 2000

- Modified the prior data protection act for public organisations in the UK to provide the public with more rights
- Rights
 - To view the data on them held by an organisation for a reasonable fee.
 - To request that incorrect information be corrected.
 - To require that their data was not used in any way that potentially could have caused damage or distress.
 - To require that their data was not used for direct marketing.



Laws:

EU General Data Protection Regulation (GDPR)

- Again, designed to protect personal data
 - “Personal data is information that relates to an identified or identifiable individual.”
 - “You should take into account the information you are processing together with all the means reasonably likely to be used by either you or any other person to identify that individual.”
- Personal data may not be processed unless:
 - The data subject has given (explicit) consent
 - To fulfill contractual obligations with a data subject
 - To comply with a data controller's legal obligations
 - To protect the vital interests of a data subject or another individual
 - To perform a task in the public interest or in official authority
 - For the legitimate interests of a data controller or a third party
- Defines a **Data Controller** (Organisation)
 - Must declare the lawful basis and purpose for data processing, and state how long data is being retained and if it is being shared with any third parties or outside of the EEA
 - Must have a designated **Data Protection Officer** who must be contactable



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